Programming Assignment HW5

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Problem-1

(Your Codes)

RoomDimension.java

```java
package q1;

/**
 * The RoomDimension class stores the length and width of a room.
 * It can calculate and return the area of the room in square feet.
 */
public class RoomDimension {

    // Stores the length of the room.
    private double length;

    // Stores the width of the room.
    private double width;

    /**
     * Constructs a RoomDimension object with the given length and width.
     *
     * @param len the length of the room
     * @param w the width of the room
     */
    public RoomDimension(double len, double w) {
        length = len;
        width = w;
    }

    /**
     * Calculates and returns the area of the room.
     *
     * @return the area of the room in square feet
     */
    public double getArea() {
        return length * width;
    }

    /**
     * Returns a string representation of the room dimensions.
     *
     * @return a string containing the room length and width
     */
    @Override
    public String toString() {
        return "Length: " + length + " ft, Width: " + width + " ft";
    }
}
```

RoomCarpet.java

```java
package q1;

/**
 * The RoomCarpet class stores information about carpeting for a room.
 * It contains a RoomDimension object and the carpet cost per square foot.
 * It can calculate the total cost of carpeting the room.
 */
public class RoomCarpet {

    // Stores the dimensions of the room.
    private RoomDimension size;

    // Stores the carpet cost per square foot.
    private double carpetCost;

    /**
     * Constructs a RoomCarpet object with room dimensions and carpet cost.
     *
     * @param dim the RoomDimension object for the room
     * @param cost the carpet cost per square foot
     */
    public RoomCarpet(RoomDimension dim, double cost) {
        size = dim;
        carpetCost = cost;
    }

    /**
     * Calculates and returns the total carpet cost for the room.
     *
     * @return the total cost of the carpet
     */
    public double getTotalCost() {
        return size.getArea() * carpetCost;
    }

    /**
     * Returns a string representation of the carpet information.
     *
     * @return a string containing room details, carpet cost, and total cost
     */
    @Override
    public String toString() {
        return size.toString()
                + "\nCarpet Cost per Square Foot: $" + String.format("%.2f",
carpetCost)
                + "\nTotal Carpet Cost: $" + String.format("%.2f",
getTotalCost());}
    }
```

Question1Demo.java

```java
import q1.RoomCarpet;
import q1.RoomDimension;

import java.util.Scanner;

/**
 * The Question1Demo class is the driver program for the carpet cost
application.
 * It asks the user for room dimensions and carpet price, then displays total
cost.
 */
public class Question1Demo {

    /**
     * The main method gets user input, creates objects, and displays
results.
     *
     * @param args command-line arguments
     */
    public static void main(String[] args) {
        Scanner keyboard = new Scanner(System.in);

        double length;
        double width;
        double costPerSquareFoot;

        System.out.print("Enter the length of the room in feet: ");
        length = keyboard.nextDouble();

        System.out.print("Enter the width of the room in feet: ");
        width = keyboard.nextDouble();

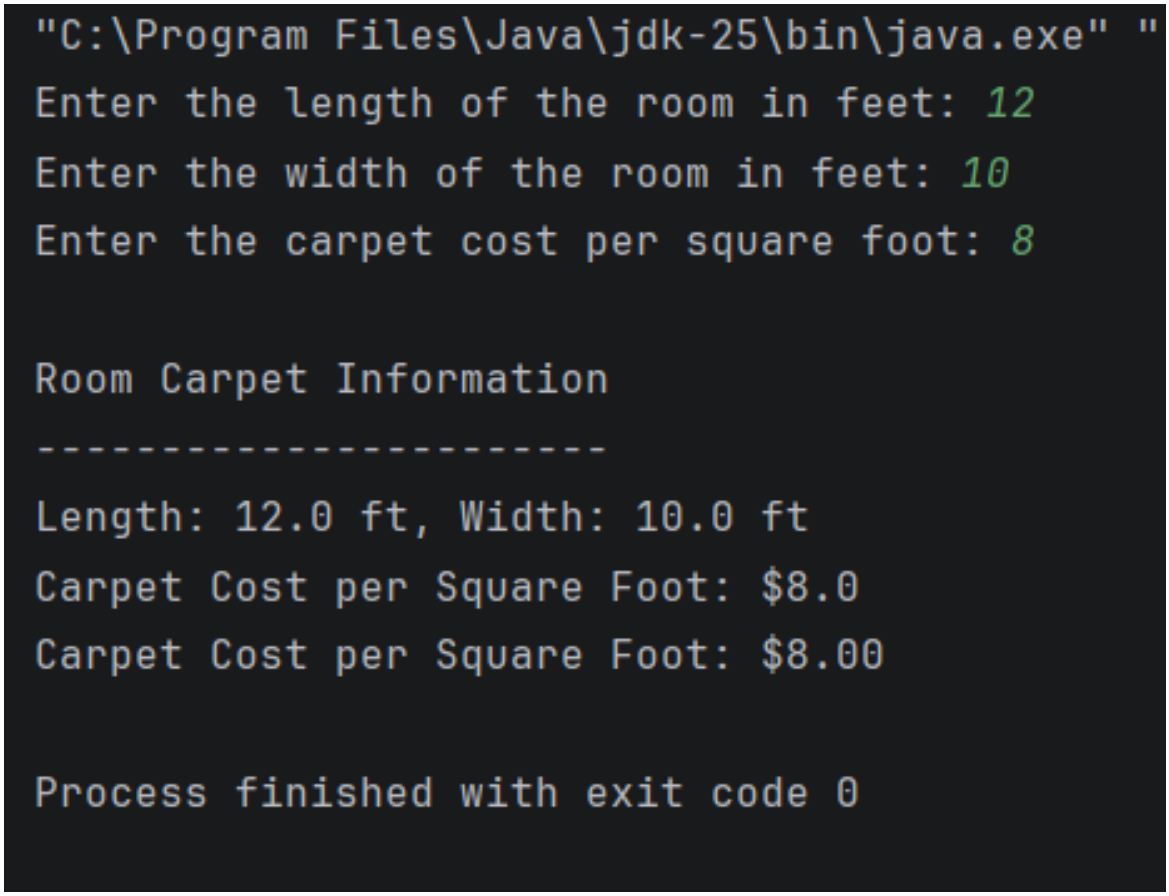
        System.out.print("Enter the carpet cost per square foot: ");
        costPerSquareFoot = keyboard.nextDouble();

        RoomDimension room = new RoomDimension(length, width);
        RoomCarpet carpet = new RoomCarpet(room, costPerSquareFoot);

        System.out.println("\nRoom Carpet Information");
        System.out.println("-----------------------");
        System.out.println(carpet);

        keyboard.close();
    }
}
```

(Output- screenshots)



Problem-2

(Your Codes)

Month.java

```java
package q2;

/**
 * The Month class stores the number of a month and provides
 * methods to get the month name and compare months.
 */
public class Month {

    // Stores the number of the month.
    private int monthNumber;

    /**
     * No-arg constructor sets monthNumber to 1.
     */
    public Month() {
        monthNumber = 1;
    }

    /**
     * Constructor that accepts a month number.
     * If invalid, sets to 1.
     *
     * @param month the month number
     */
    public Month(int month) {
        if (month >= 1 && month <= 12) {
            monthNumber = month;
        } else {
            monthNumber = 1;
        }
    }

    /**
     * Constructor that accepts a month name.
     *
     * @param name the month name
     */
    public Month(String name) {
        setMonthFromName(name);
    }

    /**
     * Sets the month number.
     *
     * @param month the month number
     */
    public void setMonthNumber(int month) {
        if (month >= 1 && month <= 12) {
            monthNumber = month;
        } else {
```

```
            monthNumber = 1;
        }
    }

    /**
     * Returns the month number.
     *
     * @return month number
     */
    public int getMonthNumber() {
        return monthNumber;
    }

    /**
     * Returns the name of the month.
     *
     * @return month name
     */
    public String getMonthName() {
        String[] months = {
                "January", "February", "March", "April",
                "May", "June", "July", "August",
                "September", "October", "November", "December"
        };
        return months[monthNumber - 1];
    }

    /**
     * Returns the month name.
     *
     * @return month name
     */
    @Override
    public String toString() {
        return getMonthName();
    }

    /**
     * Checks if months are equal.
     *
     * @param other another Month object
     * @return true if equal
     */
    public boolean equals(Month other) {
        return this.monthNumber == other.monthNumber;
    }

    /**
     * Checks if this month is greater.
     *
     * @param other another Month object
     * @return true if greater
     */
    public boolean greaterThan(Month other) {
        return this.monthNumber > other.monthNumber;
    }
```

```java
    /**
     * Checks if this month is less.
     *
     * @param other another Month object
     * @return true if less
     */
    public boolean lessThan(Month other) {
        return this.monthNumber < other.monthNumber;
    }

    /**
     * Converts a month name to its number.
     *
     * @param name the month name
     */
    private void setMonthFromName(String name) {
        String[] months = {
                "January", "February", "March", "April",
                "May", "June", "July", "August",
                "September", "October", "November", "December"
        };

        for (int i = 0; i < months.length; i++) {
            if (months[i].equalsIgnoreCase(name)) {
                monthNumber = i + 1;
                return;
            }
        }

        monthNumber = 1;
    }
}
```

Question2Demo.java

```java
package q2;

/**
 * Demonstrates the Month class.
 */
public class Question2Demo {

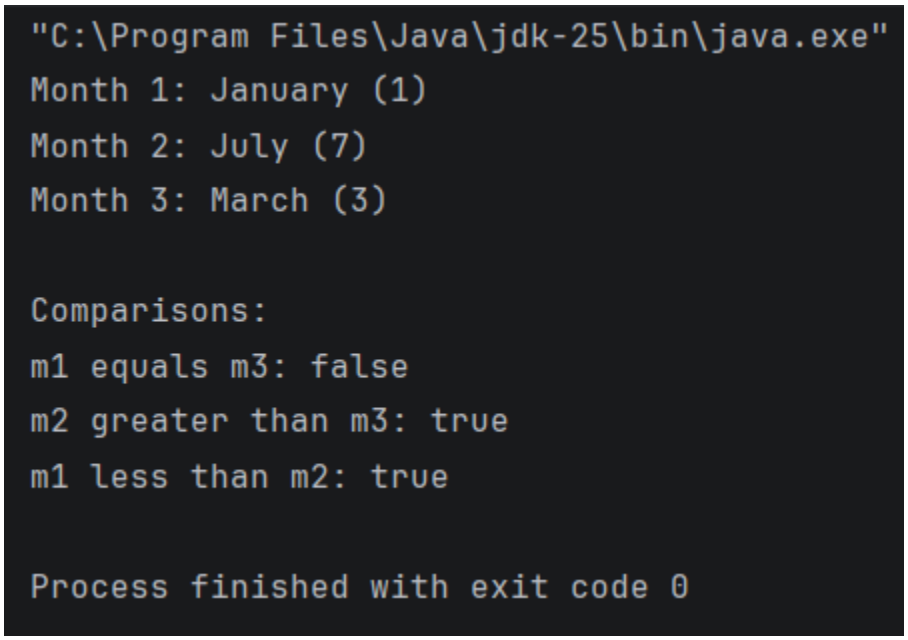
    /**
     * Main method to test Month class.
     *
     * @param args command-line arguments
     */
    public static void main(String[] args) {

        Month m1 = new Month(1);          // January
        Month m2 = new Month(7);          // July
        Month m3 = new Month("March");    // March

        System.out.println("Month 1: " + m1 + " (" + m1.getMonthNumber() +
")");
        System.out.println("Month 2: " + m2 + " (" + m2.getMonthNumber() +
")");
        System.out.println("Month 3: " + m3 + " (" + m3.getMonthNumber() +
")");

        System.out.println("\nComparisons:");
        System.out.println("m1 equals m3: " + m1.equals(m3));
        System.out.println("m2 greater than m3: " + m2.greaterThan(m3));
        System.out.println("m1 less than m2: " + m1.lessThan(m2));
    }
}
```

(Output- screenshots)



Problem-3

(Your Codes)

Date.java

```java
package q3;

import q2.Month;

/**
 * The Date class represents an immutable date.
 * It stores a month, day, and year and validates input values.
 */
public final class Date {

    // Stores the month using the Month class.
    private final Month month;

    // Stores the day of the month.
    private final int day;

    // Stores the year.
    private final int year;

    /**
     * Constructor that accepts month, day, and year.
     * Validates all values.
     *
     * @param m the month (1-12)
     * @param d the day
     * @param y the year
     */
    public Date(int m, int d, int y) {

        // Validate month
        if (m < 1 || m > 12) {
            m = 1;
        }

        this.month = new Month(m);

        // Validate year (cannot be 0)
        if (y == 0) {
            y = 1;
        }

        this.year = y;

        // Validate day based on month
        int maxDays = getDaysInMonth(m);

        if (d < 1 || d > maxDays) {
            d = 1;
        }
```

```
        this.day = d;
    }

    /**
     * Copy constructor.
     *
     * @param other another Date object
     */
    public Date(Date other) {
        this.month = new Month(other.month.getMonthNumber());
        this.day = other.day;
        this.year = other.year;
    }

    /**
     * Returns the number of days in a given month.
     *
     * @param m the month number
     * @return number of days
     */
    private int getDaysInMonth(int m) {
        switch (m) {
            case 2:
                return 28;
            case 4: case 6: case 9: case 11:
                return 30;
            default:
                return 31;
        }
    }

    /**
     * Returns the date as a formatted string.
     *
     * @return formatted date string
     */
    @Override
    public String toString() {
        return month.getMonthName() + " " + day + ", " + year;
    }
}
```

Question3Demo.java

```
package q3;

import java.util.Scanner;

/**
 * Demonstrates the Date class using user input.
 */
public class Question3Demo {

    /**
     * Main method asks user for a date and displays it.
     *
     * @param args command-line arguments
     */
```

```
    public static void main(String[] args) {

        Scanner keyboard = new Scanner(System.in);

        System.out.print("Enter month (1-12): ");
        int month = keyboard.nextInt();

        System.out.print("Enter day: ");
        int day = keyboard.nextInt();

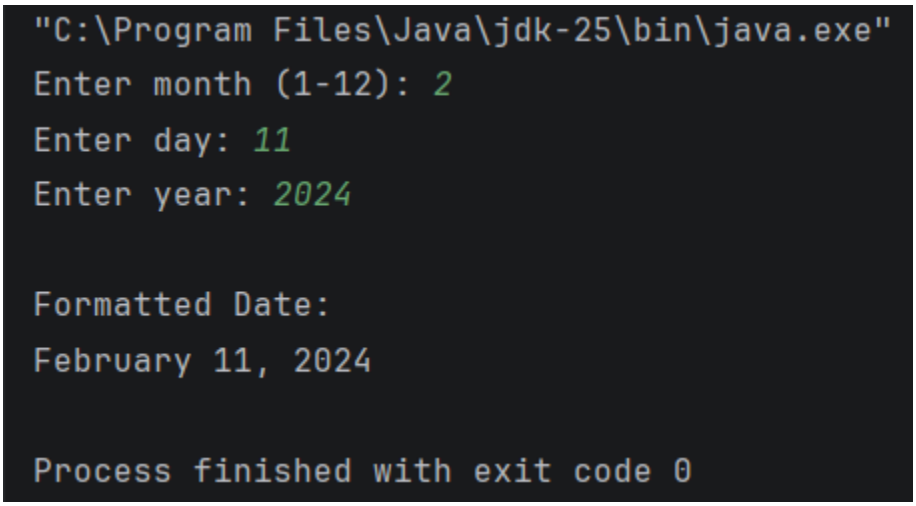
        System.out.print("Enter year: ");
        int year = keyboard.nextInt();

        Date date = new Date(month, day, year);

        System.out.println("\nFormatted Date:");
        System.out.println(date);

        keyboard.close();
    }
}
```

(Output- screenshots)



Problem-4

(UML)

